

PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	A01

U.S. DEPARTMENT OF THE INTERIOR
U.S. FISH AND WILDLIFE SERVICE
SANDY POINT NATIONAL
WILDLIFE REFUGE

PLANS FOR PROPOSED PROJECT

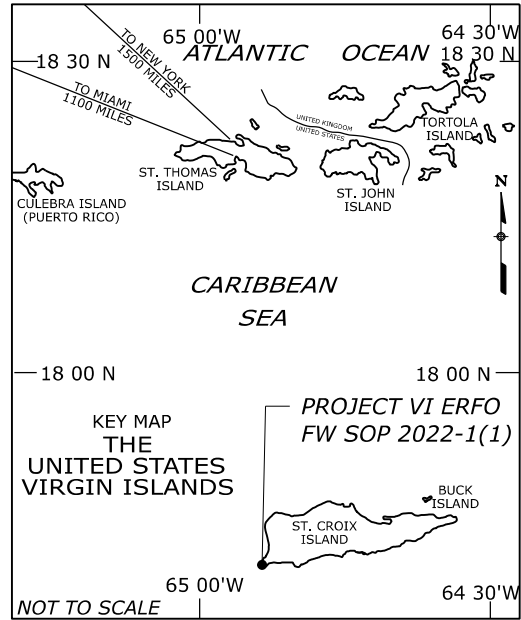
VI ERFO FW SOP 2022-1(1)

WOODEN BRIDGE REPLACEMENT

THE ISLAND OF ST. CROIX
U.S. VIRGIN ISLANDS

INDEX TO SHEETS

SHEET NO	DESCRIPTION
A01	Title Sheet
A02-A03	Symbols And Abbreviations
A04	Location Map and Temporary Traffic Control Plan
C01	Tabulation of Quantities
M01	Erosion and Sediment Control Narrative
M02	Erosion and Sediment Control Plan
R01-R05	Bridge Plans
S01	Standards And Details



DESCRIPTION OF PROJECT

IMPROVEMENT: Replacement of flood damaged wooden bridge from Hurricane Fiona.

PROJECT LENGTH: 0.003 Miles

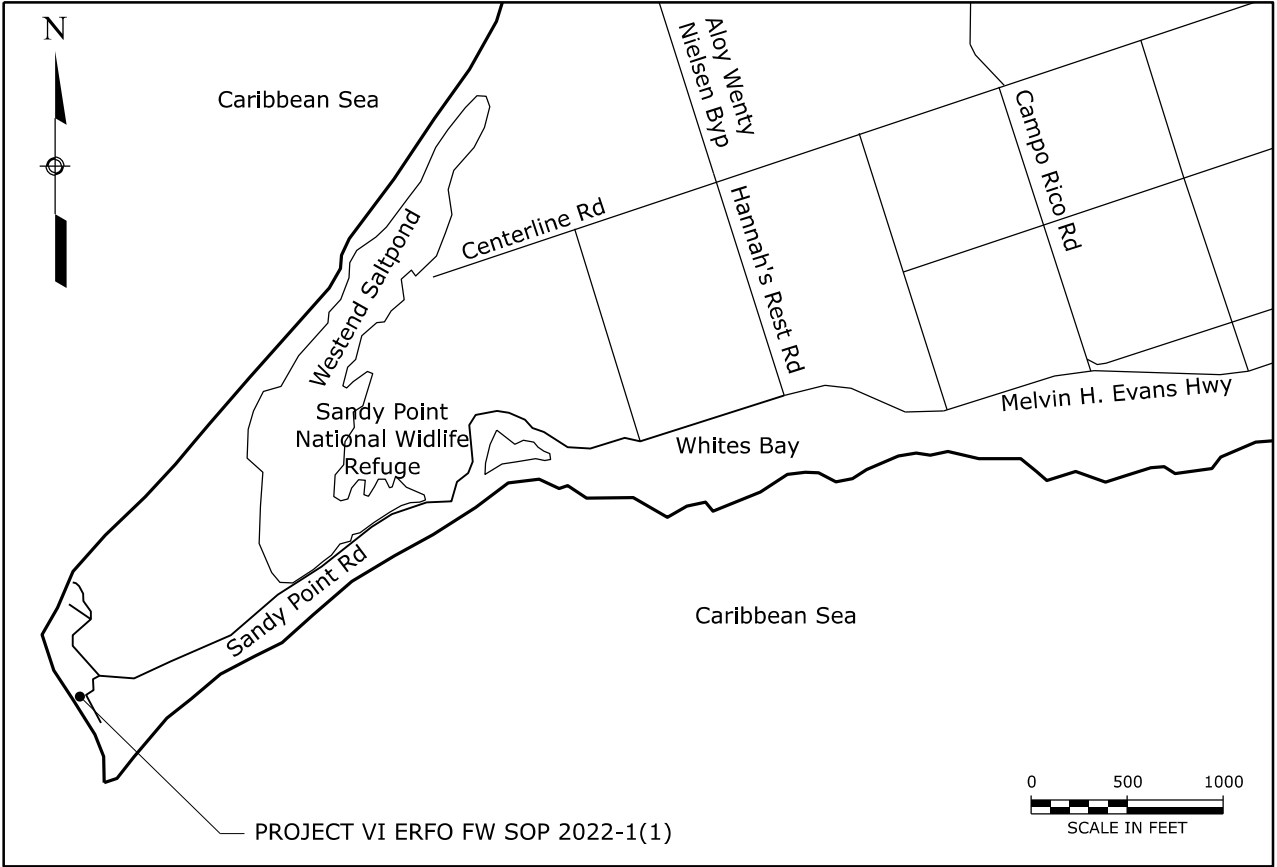
LANE MILES: N/A

BRIDGE: West Beach Parking Pedestrian Bridge

DESIGN DESIGNATION: N/A

SPECIFICATIONS:

"Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects", FP-24.



Know what's below.
Call before you dig.



PLANS PREPARED BY
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
EASTERN FEDERAL LANDS HIGHWAY DIVISION
ASHBURN, VIRGINIA
FEBRUARY, 2026

Sheet 1 of 2

PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	A03

North Arrow



Slope Stake Limits



Fence



Gate with Fence



Cattleguard



Guardrail



Concrete Barrier



Retaining Wall



Signs (single, double post; portable)



Delineators



Pipe Culvert (arrow shows flow)



Pipe Culvert with End Section



Pipe Culvert with Headwall



Pipe Culvert with Drop Inlet



Box Culvert



Underdrain



Overhead/Above Ground Utilities



Underground Utilities



FM = force main, FO = fiber optic, G = gas, IRR = irrigation, O = oil,
P = power, SA = sanitary sewer, SD = storm drain, SS = storm sewer,
STEAM = steam, T = telephone, TV = CATV, W = water

Poles (Power, Telephone, Joint Use,
Light, Support w/Anchor)



Miscellaneous Utility Features

EM = electric meter, T = telephone pedestal, TV = CATV pedestal,
UP = transformer or junction box, WF = water fountain

Building



Right-of-Way Line with Monument



Permanent Easement



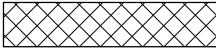
Construction Easement



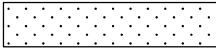
Riprap



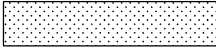
Pavement Removal / Roadway Obliteration



Full Depth Pavement



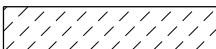
Sidewalk Asphalt/Concrete



Mill and Overlay



Overlay



Silt Fence



Diversion Berm



Drainage Divide



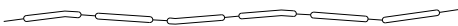
Check Dam



Limits of Disturbance



Fiber Roll or Wattle



PROJECT SPECIFIC SYMBOLS AND ABBREVIATIONS:

WETLAND BOUNDARY



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NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

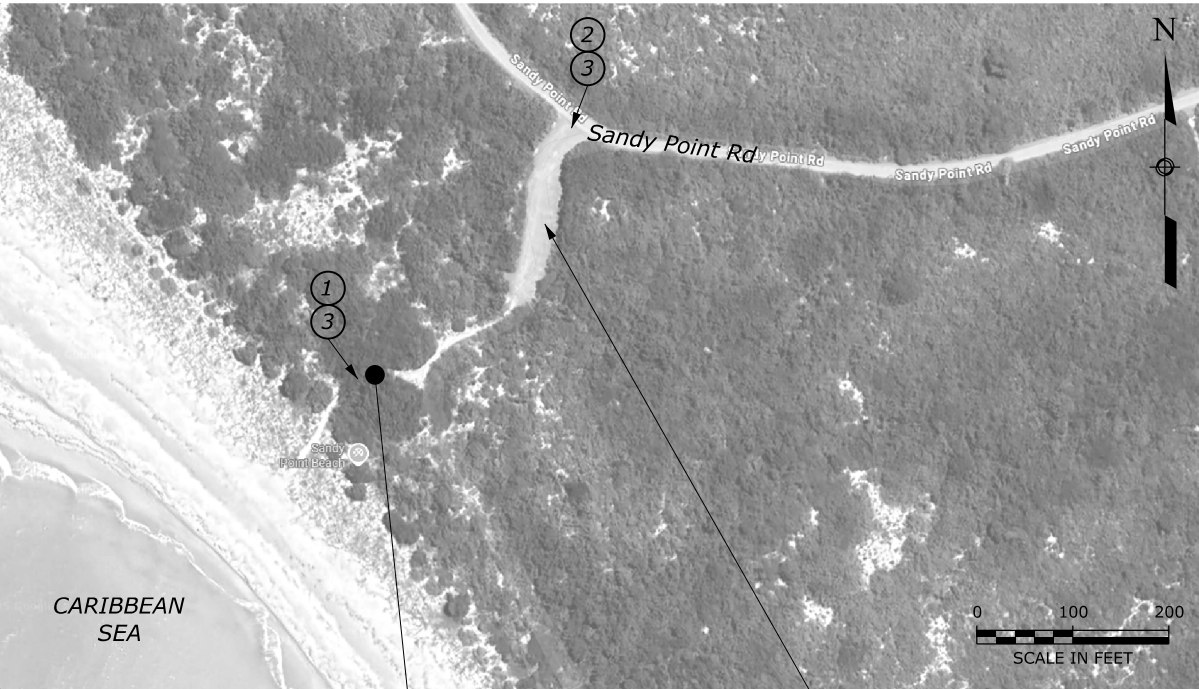
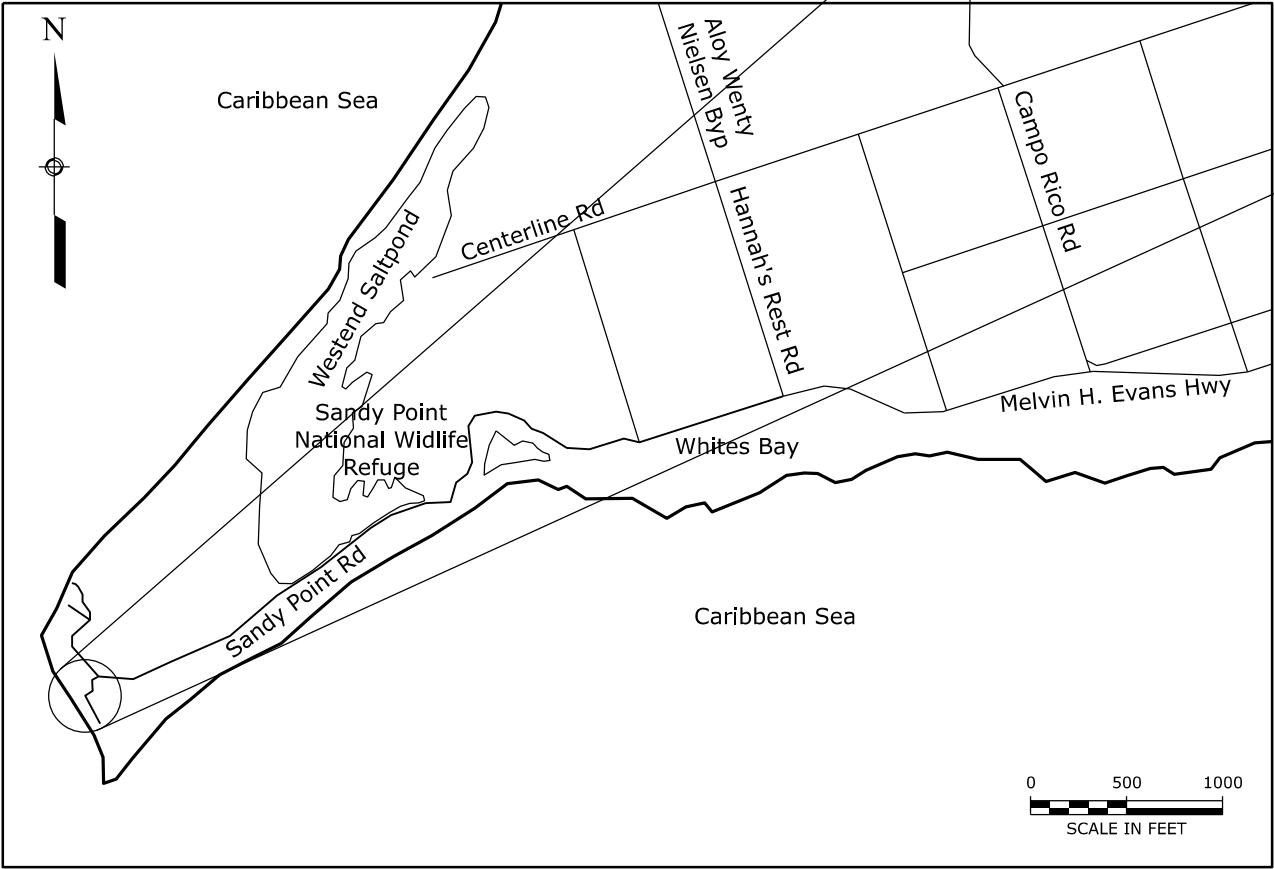
SANDY POINT NATIONAL WILDLIFE REFUGE

SYMBOLS AND ABBREVIATIONS

PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	A04

NOTE:

Place signs and Type 3 Barricades at the bridge ends. Coordinate with the CO for the final location of the barricades.



VI ERFO FW SOP 2022-1(1)
(17.677194, -64.901778)

- 1

AREA
CLOSED
EM2-1
- 2

ROAD
CLOSED
- 3

Barricade Type 3

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY

SANDY POINT NATIONAL WILDLIFE REFUGE
**LOCATION MAP AND
TEMPORARY TRAFFIC
CONTROL PLAN**

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Line Item No.	Pay Item No.	Pay Item Description	Unit	Estimated Quantities
				Bid Schedule
A1000	15101-0000	MOBILIZATION	LPSM	ALL
A1020	15201-0000	THIRD-PARTY COMPENSATION	CTSM	ALL
A1040	15201-0000	CONSTRUCTION SURVEY AND STAKING	LPSM	ALL
A1060	15401-0000	CONTRACTOR TESTING	LPSM	ALL
A1080	15705-0100	SOIL EROSION CONTROL, SILT FENCE	LNFT	32
A1100	20304-1000	REMOVAL OF STRUCTURES AND OBSTRUCTIONS	LPSM	ALL
A1120	20801-0000	STRUCTURE EXCAVATION	CUYD	3
A1140	55701-2000	STRUCTURAL TIMBER AND LUMBER, TREATED	MFBM	1
A1160	55701-3000	STRUCTURAL TIMBER AND LUMBER, COMPOSITE	MFBM	1
A1180	57302-0000	HELICAL PILE	EACH	4
A1200	63502-0600	TEMPORARY TRAFFIC CONTROL, BARRICADE TYPE 3	EACH	4
A1220	63504-1000	TEMPORARY TRAFFIC CONTROL, CONSTRUCTION SIGN	SQFT	20
A1240	63701-0000	FIELD OFFICE	EACH	1

NO.	DATE	BY	REVISIONS

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U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY

SANDY POINT NATIONAL WILDLIFE REFUGE
TABULATION OF QUANTITIES

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PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	M01

Project Description

This project consists of bridge replacement.

Soil disturbing activities include installing helical piles. The total disturbed area for the project is approximately 73.33 sqft. The receiving water is the West End Salt Pond or the Carribean Sea.

No new impervious surface will be created by the bridge replacement. The Runoff Coefficient prior to construction and after construction will remain the same. Notify the Contracting Officer (CO) as soon as possible if the disturbance exceeds 1 acre.

Prohibited Discharges

The following discharges are prohibited:

- Wash-water from concrete, paint, curing compounds, and other construction materials*
- Fuels, oils, equipment-related compounds*
- Soaps, solvents used for vehicle washing*
- Waste, garbage, sanitary waste*

Inspect and maintain on a regular basis, all mechanized equipment used in or near surface water to prevent contamination from fuels, lubricants, hydraulic fluids, or other toxic materials.

Solid waste generated from the project will consist of construction debris, garbage, and empty containers. Collect and store all waste in dumpsters, or in metal or plastic drums, as appropriate.

Hazardous waste will not be generated from normal construction activities. Equipment fueling and maintenance could generate spills, leaks, and hazardous wastes like motor oil, diesel, gasoline, and battery fluid. If feasible, conduct these activities in a covered area to avoid contact with storm water. Store all hazardous waste materials in appropriate and clearly marked containers away from other non-waste materials. Do not dispose of hazardous waste materials into the on-site dumpsters. Dispose of material according to Federal, State, and local regulations.

Develop and implement a Spill Prevention Control and Countermeasures (SPCC) plan following the requirements under 40 CFR 112. Report spills large enough to discharge to surface waters to the National Response Center at 1-800-424-8802.

General Guidelines

The Erosion & Sediment Control Narrative is meant as a guideline for preventing erosion and controlling sediment. The work consists of applying measures throughout the life of the project to control erosion and to minimize the sedimentation of rivers, streams, and impoundments such as lakes, reservoirs, bays, and coastal waters. The measures consist of soil erosion control measures which are also defined and outlined in the Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects, FP-24, and the Special Contract Requirements.

Do not modify the type, size, or location of any control or practice without prior approval from the Contracting Officer (CO).

No construction access will be permitted through a wetland or waterway.

Do not allow construction vehicles to track sediment outside the project limits.

Do not allow any construction equipment to operate on or access the down-slope side of the perimeter control measures.

Direct storm water to vegetated buffer areas and do not discharge directly into surface waters.

Sequence of Construction

Phase I Establish Perimeter Controls

Prior to any clearing, grubbing, or excavation, construct perimeter controls to ensure that disturbed sediment does not leave the project site. Perimeter controls include silt fence and other specified measures outside the construction limits.

Phase II Intermediate Controls

Apply intermediate controls during rough grading operations. Install silt fence on both sides of the bridge.

At the end of each day's grading operations, shape earthwork to minimize and control erosion from storm runoff.

Provide watering for dust control within the construction limits, on active haul roads, and in pits and staging areas.

Phase III Final Construction / Stabilization

Remove all other perimeter controls when directed by the CO.

Maintenance and Inspection Procedures

Unless stated otherwise, construct and maintain all vegetated and structural erosion control practices according to Section 157, the details shown in the plans, and the individual permitting requirements. Check and maintain erosion control measures once every 7 days and within 24 hours after a rain of 0.25 inches or more, and daily during wet weather. Repair or replace any damaged measures by the end of the day.

Silt fence - inspect for buildup of excess sediment, undercutting, sags, and other failures. If the fabric becomes damaged, repair or replace as necessary. Remove sediment from behind the silt fence when it becomes 0.5 feet deep at the fence.

Record the inspection date and summary of findings within 24 hours of completing a site inspection.

NO.	DATE	BY	REVISIONS

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

SANDY POINT NATIONAL WILDLIFE REFUGE

**EROSION AND SEDIMENT
CONTROL NARRATIVE**

PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	M02



Wetland boundary

Limit of disturbance

Proposed structure

Place silt fence



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U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
OFFICE OF FEDERAL LANDS HIGHWAY

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SCALE IN FEET

SANDY POINT NATIONAL WILDLIFE REFUGE

EROSION AND SEDIMENT
CONTROL PLAN

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2/6/2026

PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	R01

GENERAL NOTES:

SPECIFICATIONS:

Design:
AASHTO LRFD Bridge Design Specifications, 9th Edition, 2020.

Construction:
Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects, FP-24.

DESIGN LOADS:

Dead Load:
Timber: 50 pcf

Live Load:
Pedestrian: 90 psf
UTV: 1,600 lb

MATERIALS:

Structural Steel:
Provide Type 316 stainless steel for all nails, screws, strap ties, and hurricane ties.
Provide nails conforming to ASTM F1667.
Provide square drive deck screws conforming to ASTM F593.
Provide bolts, nuts, and washers galvanized according to ASTM A153.
Provide bolts conforming to ASTM A307.
Provide nuts conforming to ASTM A563.
Provide washers conforming to ASTM F436.

Timber:
All sawn lumber and wood-plastic composite callouts are nominal dimensions unless otherwise noted.
Provide pressure treated timber in accordance with AWP A Standard U1, use category UC4C. Use a waterborne copper-based suspension containing micronized or dispersed copper that does not contain arsenic or chromium compounds. Obtain all timber from a single approved source.
Provide Southern Pine No. 2 or better that conforms to the design values from AASHTO LRFD Bridge Design Specifications Table 8.4.1.1.4-1 — Reference Design Values for Visually Graded Sawn Lumber.
Provide Southern Pine structural glued laminated timber Stress Class 26F-1.9E or better that conforms to the design values from AASHTO LRFD Bridge Design Specifications Table 8.4.1.2.3-1 — Reference Design Values for Structural Glued Laminated Softwood Timber Combinations.

Welding:
Perform all welding in accordance with AASHTO/AWS D1.5 Bridge Welding Code. Perform welding using low hydrogen electrodes, classified E80XX.
Perform non-destructive testing of welds as follows: 100% visual for fillet welds.

Wood-plastic composite decking:
Submit composite decking samples for approval at least 30 days before beginning construction.
Install decking according to Section 557.

SCOPE OF WORK:

SCHEDULE A:

- Remove existing timber boardwalk.
- Excavate for substructure.
- Install helical piles.
- Install timber headers, timber rim joists, and glulam stringers.
- Install wood-plastic composite decking.
- Install curb.
- Backfill structure and grade approach ramps.

INDEX OF BRIDGE PLAN SHEETS

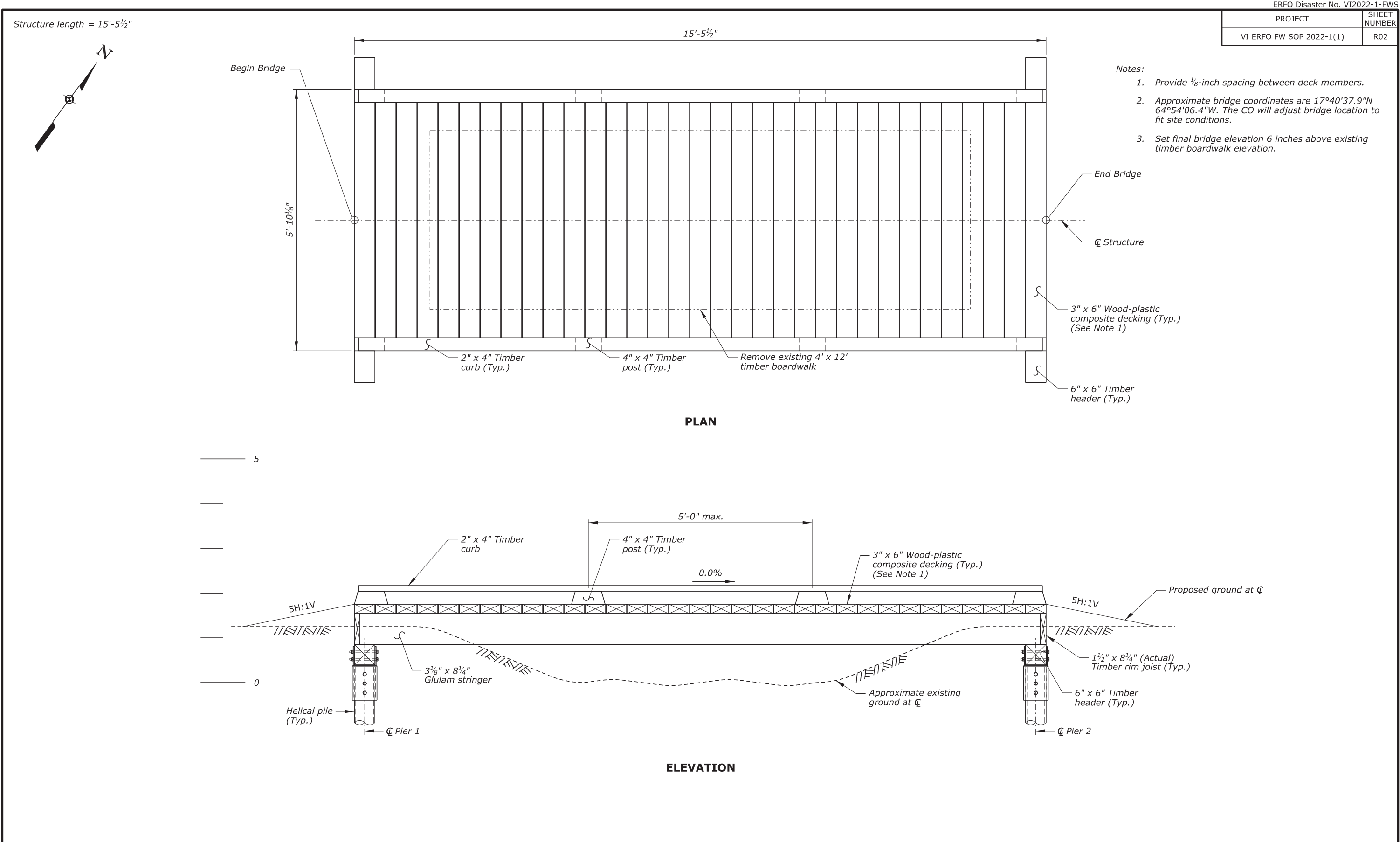
SHEET NO.	SHEET TITLE
1	General Notes and Index
2	Plan and Elevation
3	Framing Plan
4	Curb Details
5	Helical Pile Details

LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
Sandy Point National Wildlife Refuge				
PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
Christopher Negley	Not to Scale	1 of 5	February 2026	BRP-1369

GENERAL NOTES AND INDEX

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2/6/2026



STRUCTURE NO.			STRUCTURE NAME	FEATURES INTERSECTED			LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
			West Branch Parking Pedestrian Bridge				Sandy Point National Wildlife Refuge				
NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
				DJL	DJL	MRB	Christopher Negley	1" = 1'-0"	2 of 5	February 2026	BRP-1369

PLAN AND ELEVATION

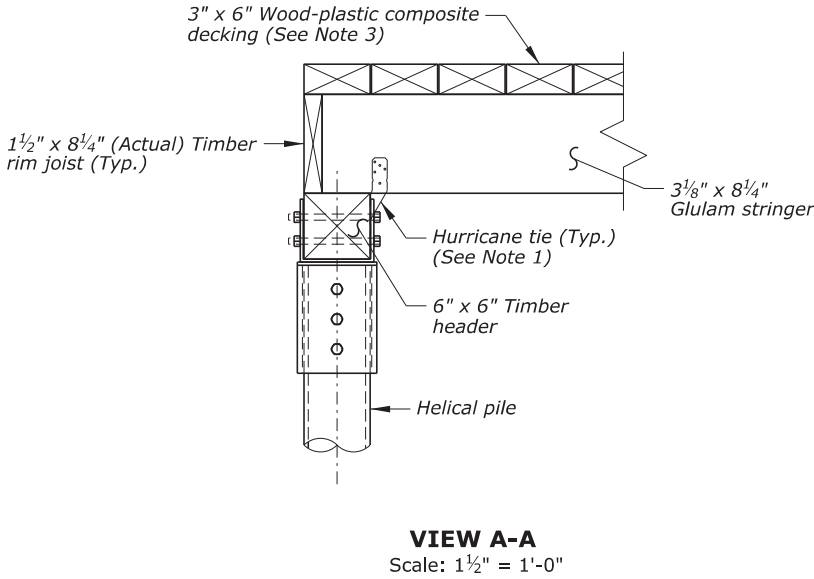
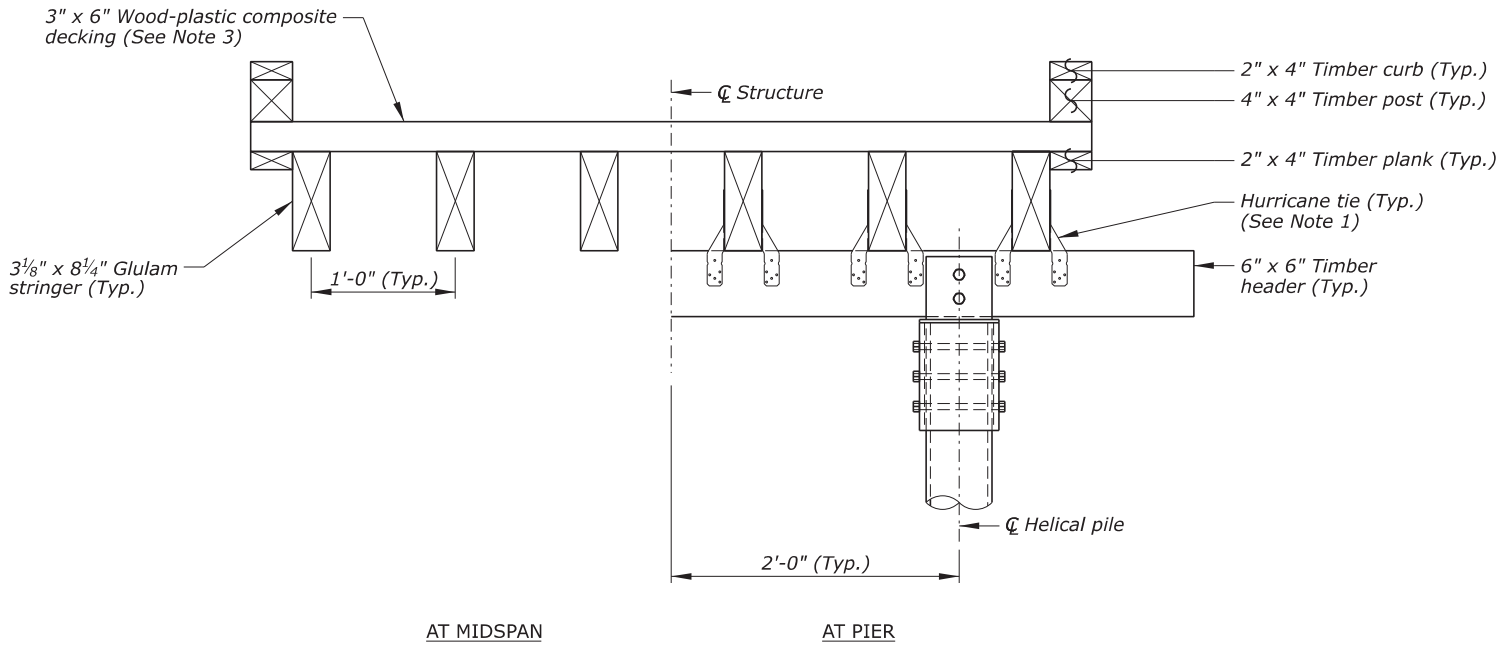
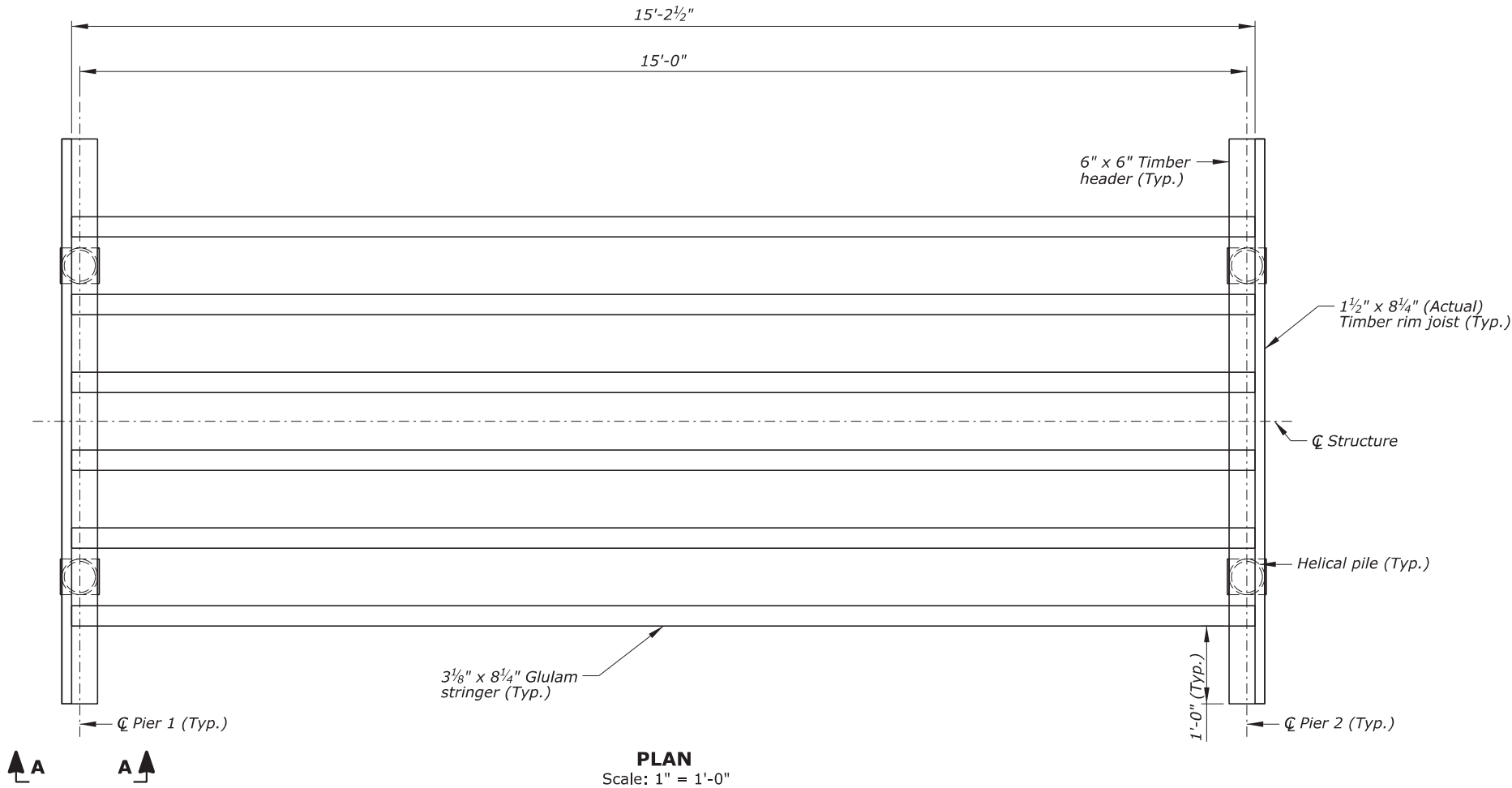
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2/6/2026

PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	R03

Notes:

1. Attach stringers to headers with stainless steel hurricane ties. Attach according to the manufacturer's recommendations, as approved.
2. Submit self-adhering flashing tape for approval at least 30 days before beginning construction. Provide self-adhering flashing tape conforming to Subsection 725.21. Install self-adhering flashing tape on the top surface of all headers, stringers, and rim joists according to the manufacturer's recommendations.
3. Provide 1/8-inch spacing between deck members.
4. Connect decking to stringers with at least two 4-inch long #10 deck screws at each support location. Do not install deck screws less than 3/4 inches away from the edge of either member. Provide pilot holes for deck screws as recommended by the manufacturer of the elements being connected.
5. Connect rim joists to stringers with at least two 16d nails at each support location. Connect rim joists to headers with 5 strap ties evenly spaced along the member. Provide 1-inch wide, 8-inch long, and 16 gauge thick strap ties.



STRUCTURE NO.			STRUCTURE NAME	FEATURES INTERSECTED			LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
			West Branch Parking Pedestrian Bridge				Sandy Point National Wildlife Refuge				
NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
				DJL	DJL	MRB	Christopher Negley	As Shown	3 of 5	February 2026	BRP-1369

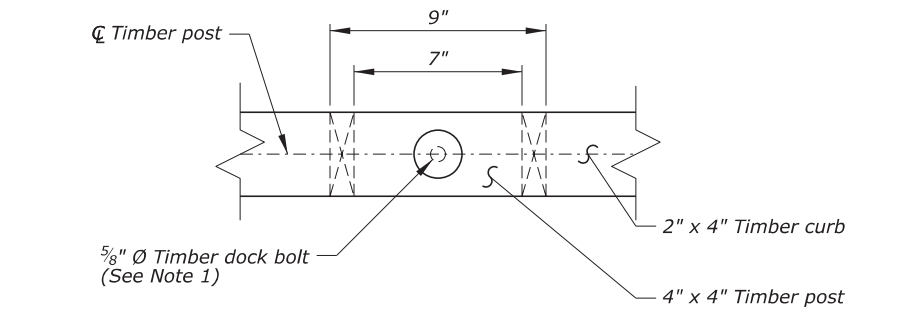
FRAMING PLAN

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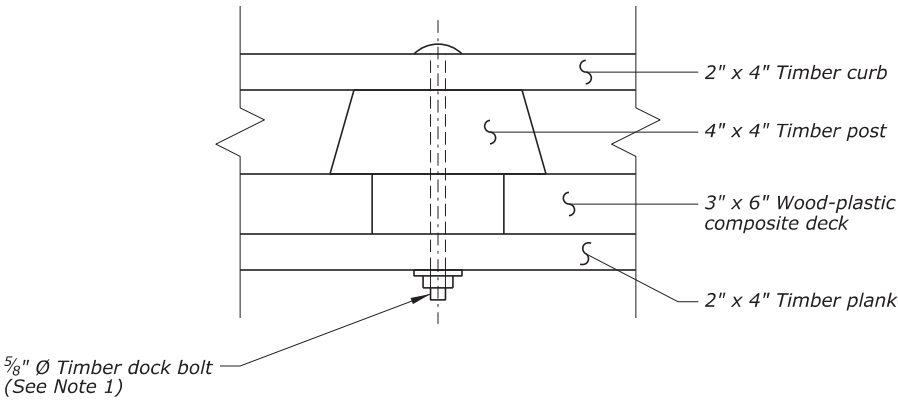
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ERFO Disaster No. VI2022-1-FWS	
PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	R04

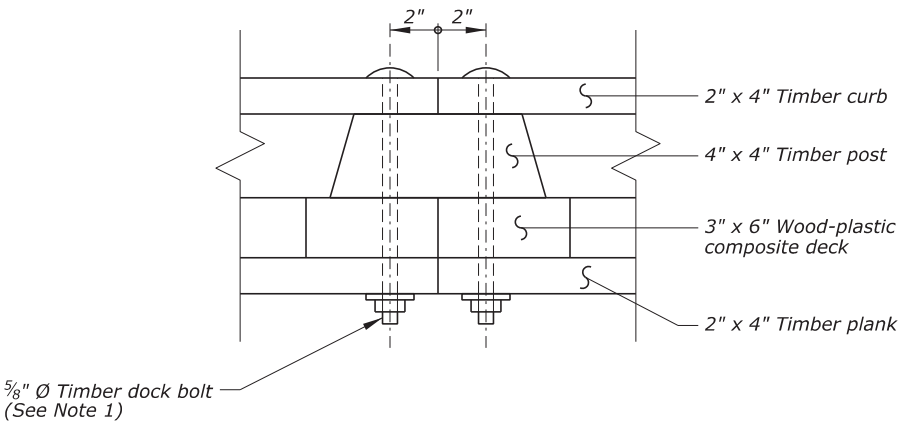
- Notes:
1. Use $\frac{5}{8}$ -inch diameter timber dock bolts with dome head, washer, and hex nut. Bore $\frac{11}{16}$ -inch diameter hole through curb, post, deck, and planks. Extend bolts $\frac{1}{2}$ -inch beyond end of nut.



PLAN VIEW



CONTINUOUS CONNECTION



DISCONTINUOUS CONNECTION

STRUCTURE NO.			STRUCTURE NAME	FEATURES INTERSECTED			LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
			West Branch Parking Pedestrian Bridge				Sandy Point National Wildlife Refuge				
NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	PROJECT TEAM LEAD	SCALE	BRIDGE PLAN SHEET	DATE	RG / BRP NO.
				DJL	DJL	MRB	Christopher Negley	3" = 1'-0"	4 of 5	February 2026	BRP-1369

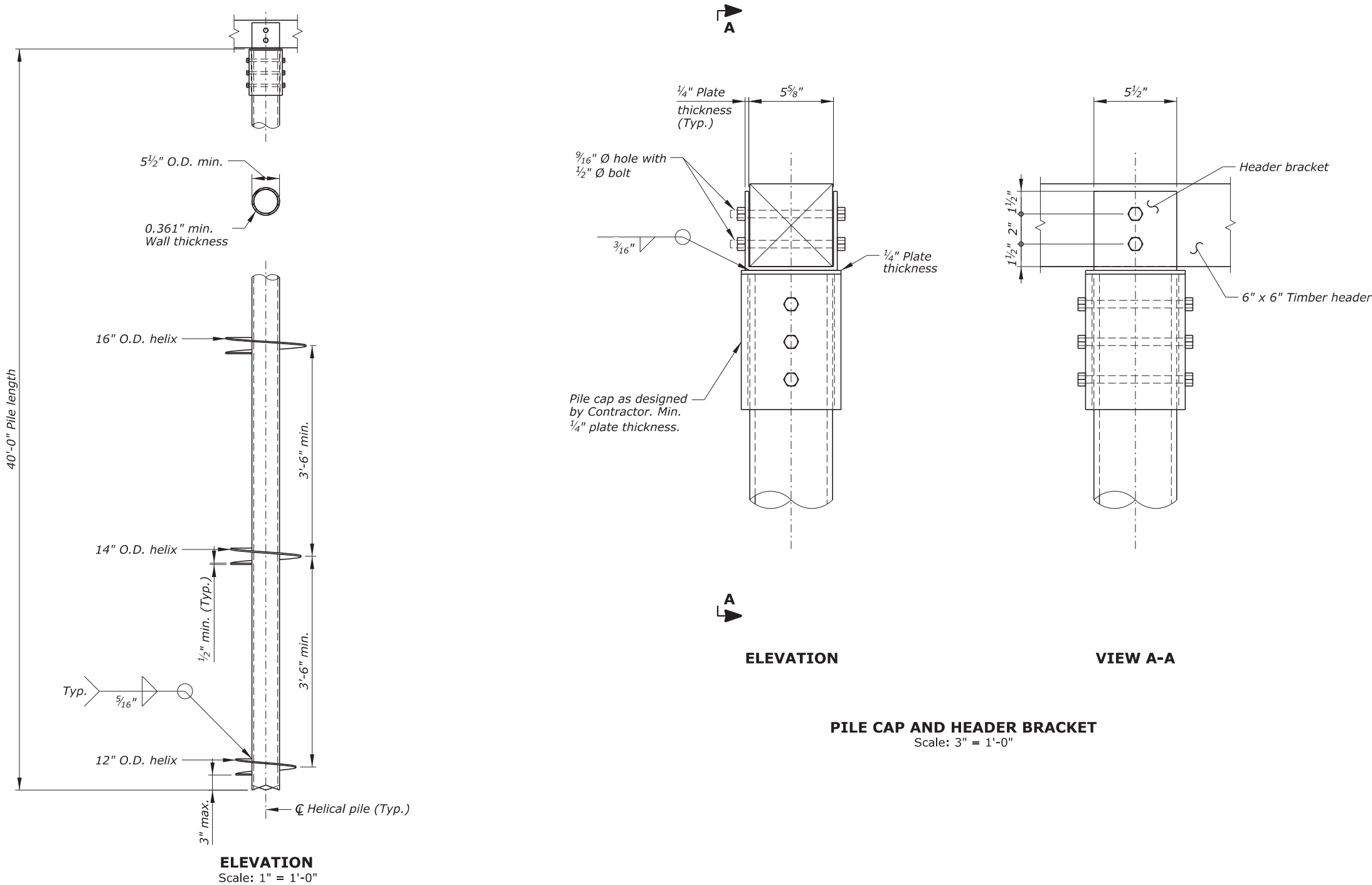
CURB DETAILS

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2/6/2026

ERFO Disaster No. VI2022-1-FWS	
PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	R05

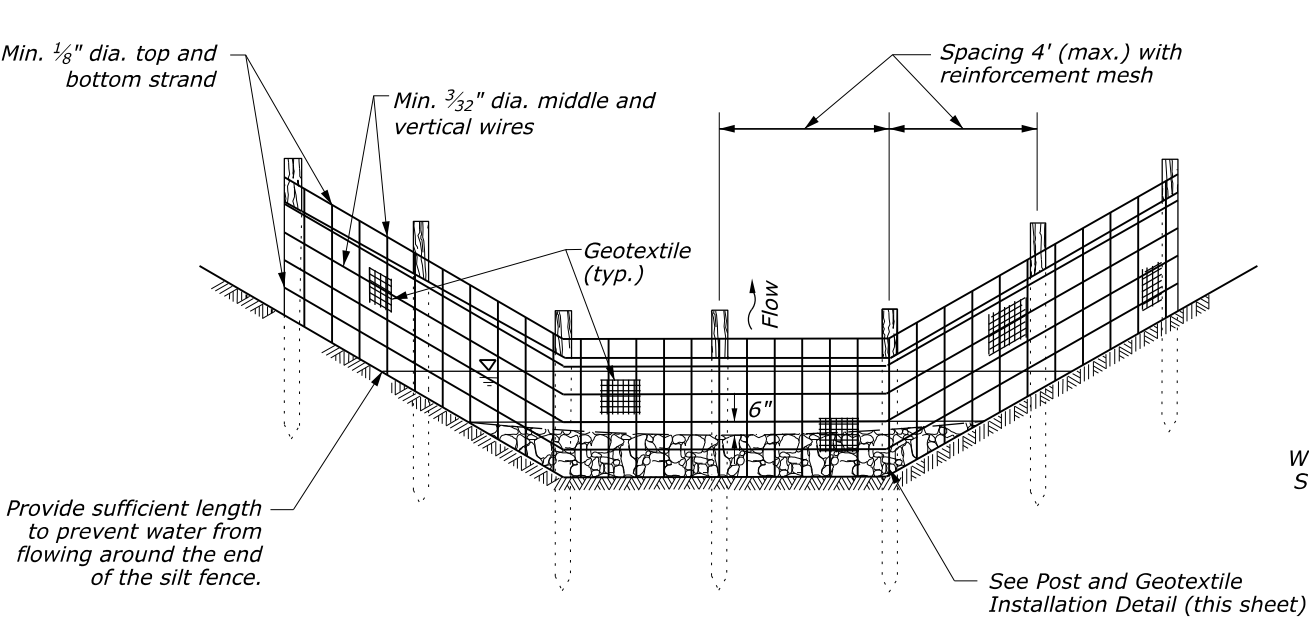
- Notes:
1. Provide high strength low alloy structural steel header brackets conforming to ASTM A572 Grade 50. Use hot-dip galvanized coating conforming to AASHTO M 111 with a minimum thickness of 3.9 mils.
 2. Provide shafts and pile caps of 80 ksi minimum yield strength.
 3. Provide helices of 50 ksi minimum yield strength.
 4. Do not use stainless steel fasteners on helical piles, pile caps, or header brackets.
 5. The maximum factored axial pile load is 4.7 kip. The maximum factored lateral pile load is 0.8 kip.



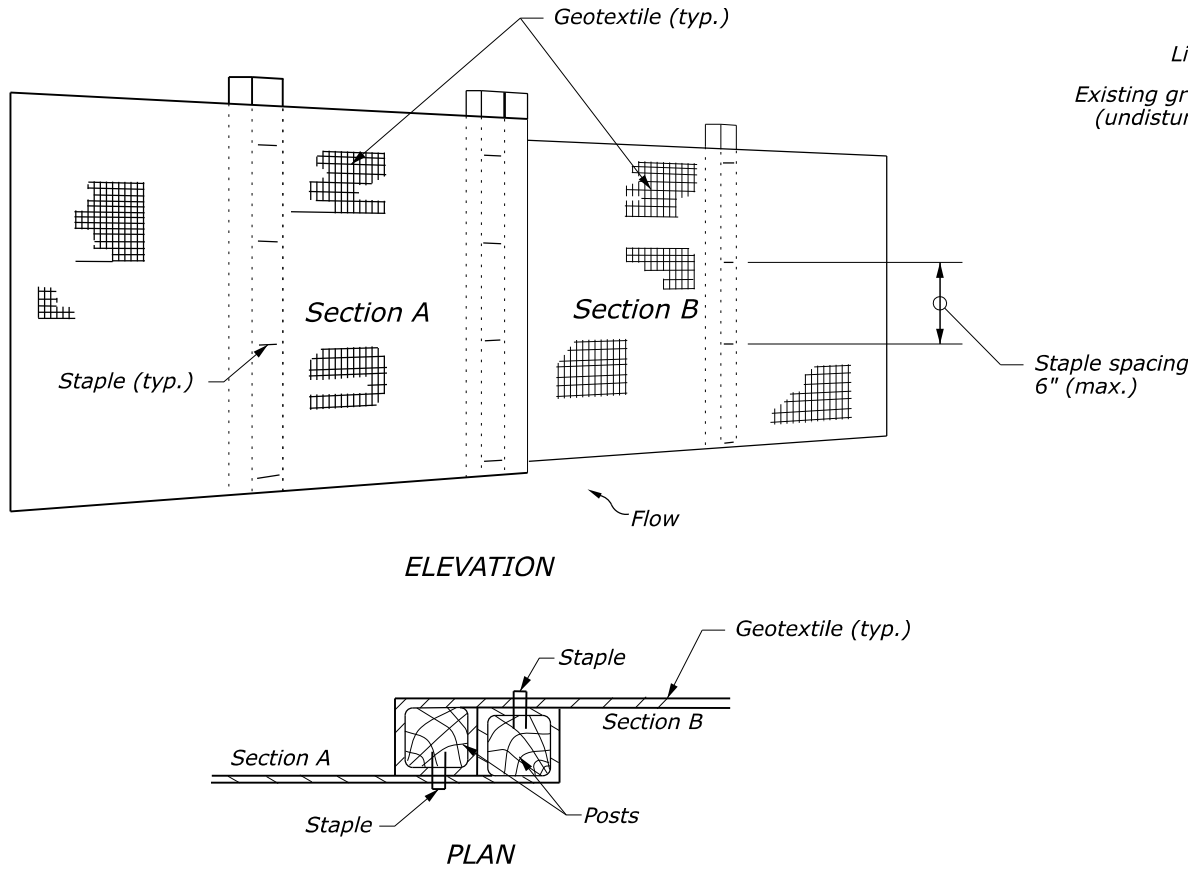
STRUCTURE NO.			STRUCTURE NAME	FEATURES INTERSECTED			LOCATION			U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
			West Branch Parking Pedestrian Bridge				Sandy Point National Wildlife Refuge				
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				DJL	DJL	MRB	Christopher Negley	As Shown	5 of 5	February 2026	BRP-1369

HELICAL PILE DETAILS

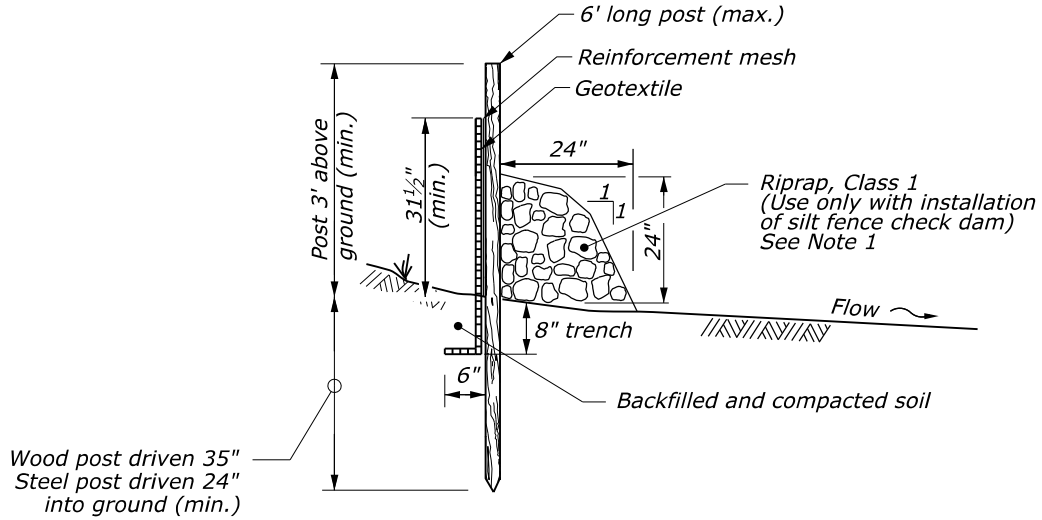
PROJECT	SHEET NUMBER
VI ERFO FW SOP 2022-1(1)	S01



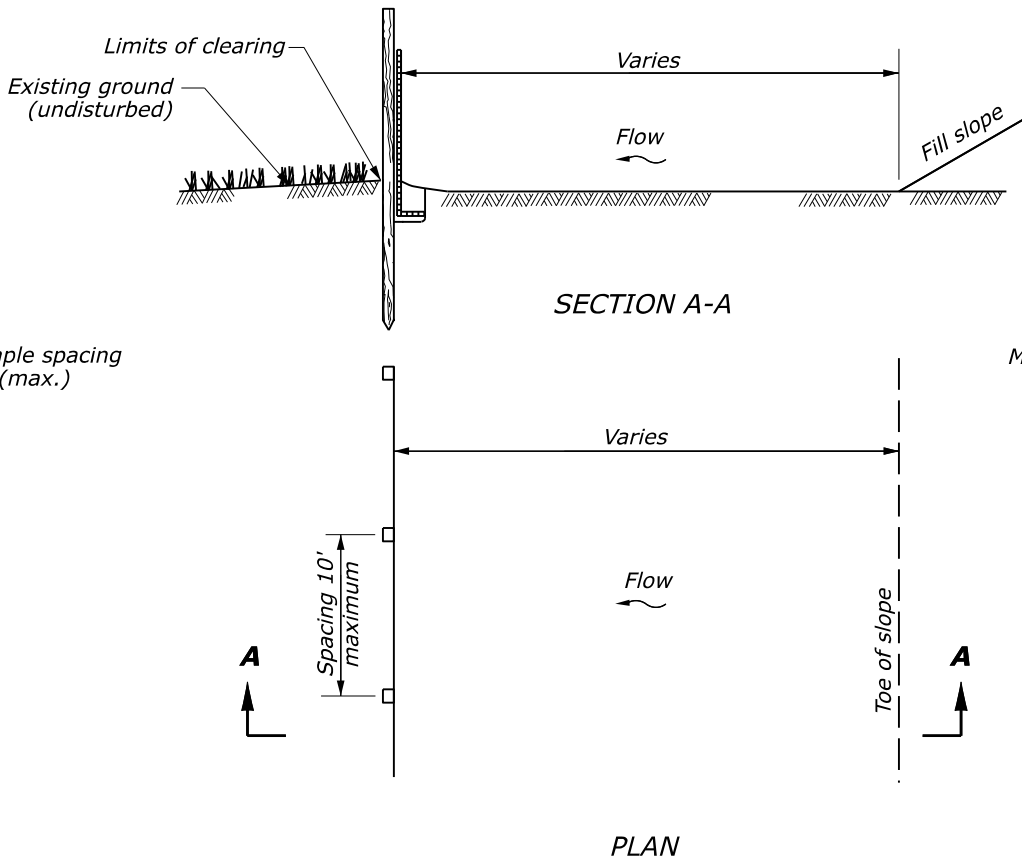
WIRE-BACKED SILT FENCE WITH CHECK DAM



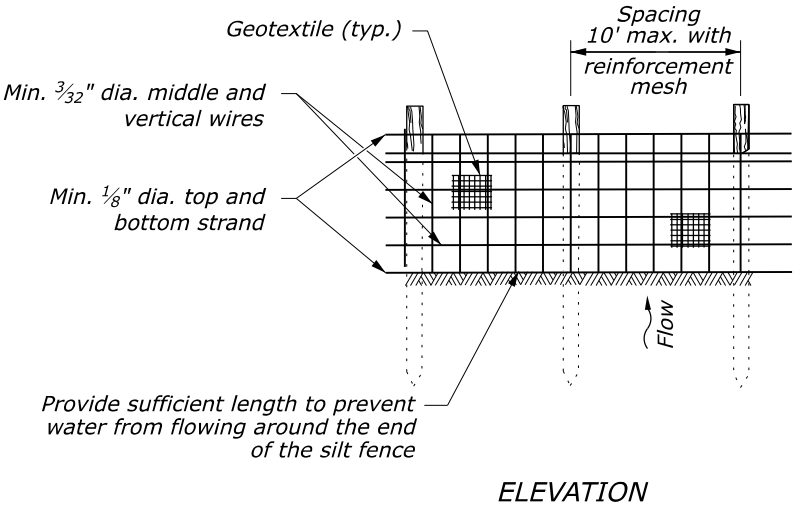
JOINING TWO ADJACENT SILT FENCE SECTIONS
(See Note 4)



POST AND GEOTEXTILE INSTALLATION



WIRE-BACKED SILT FENCE
INSTALLATION AT TOE OF FILL



NOTES:

1. Install silt fence with Class 1 riprap only in low-flow drainage ditches where shown on the Erosion and Sediment Control Plan.
2. Install pre-assembled silt fence according to the manufacturer's recommendations.
3. Install silt fence along ground contours. Curve ends of silt fence upgrade to prevent water from running around the ends.
4. Attach geotextile and reinforcement mesh so they do not slide down posts. Provide detail for attaching to steel posts for approval.
5. Use reinforcement mesh that is a minimum of 32 inches in width and has a minimum of 6 line wires with 12-inch stay spacing.
6. Use geotextile that is a minimum of 45 inches in width and fasten adequately to the reinforcement mesh as directed.
7. Use 60-inch minimum height steel posts of the self-fastener angle steel type.
8. Use 70-inch minimum height by 3-inch diameter wood posts.
9. Extend reinforcement mesh and geotextile into trench.

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA OFFICE OF FEDERAL LANDS HIGHWAY	EFLHD DETAIL E157-02
WIRE-BACKED SILT FENCE	SPECIFICATION FP-24, FP-14 APPROVED FOR USE 05/2024